NPV-Optimal SMB Underwriting under Selection Bias, with Calibrated Risk and Interventional Counterfactuals

Team Vector Ventures

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1 Problem framing & assumptions violated

We are re-underwriting a historical SMB loan book to maximize realized portfolio net present value (NPV), not to maximize classification accuracy. Several standard modelling assumptions fail, and each failure changed our approach.

Labels are missing-not-at-random behind a sharp gate. Repayment outcomes exist only for loans the prior lender approved, and approval is, to numerical precision, `prior_underwriter_score` ≥ 0.273: a propensity model separates approved from declined with AUC 1.00 and *zero* score overlap. The training label is therefore selected on a deterministic function of an observed covariate—textbook sample-selection bias. Crucially, inverse-propensity reweighting is *infeasible*: a propensity model fit on the ordinary covariates (no selection features) still reaches AUC 1.00, so approved and declined populations do not overlap in feature space and positivity is violated. Only local extrapolation near the cutoff is available; we treat PD on declined applicants as honest extrapolation and widen its intervals.

The split is forward in time. Training spans an 18-month book that ends the day before the 13-week scoring window opens, over which the default rate drifts 17.5% → 20.6%. We learn default *timing* on the history and recalibrate the *level* on the in-window validation set.

Missingness is information, not absence. "Never declined elsewhere" (null) loans default at 14.3% versus 21.3% when present; no-bank-feed loans at 19.1% versus 16.7%. We encode these as first-class indicators rather than imputing them away. **Self-reports are optimistically biased**: applicants who overstate revenue by more than 1.5× default at 38.6% versus 16.1%. The *gap* between stated and bank-feed revenue is the signal—central to our causal treatment of `stated_annual_revenue` below.

2 Methodology

One shared discrete-time model. The brief's NPV depends on the *day* of default (a day-5 default loses roughly the principal; a day-55 default nearly breaks even), and Deliverable B is the cumulative-default curve. Both consume a single timing model. We write each loan's cumulative-default curve as $F_i(t) = \mathrm{PD}_i \cdot S_{b(i)}(t)$, where $\mathrm{PD}_i = \Pr(y_i{=}1 \mid \mathbf{x}_i)$ is a calibrated probability of default by day 90 and $S_b(t)$ is a normalized timing shape ($S_b(90){=}1$) learned per credit band b.

PD model. A five-fold `GroupKFold` (by `business_id`) LightGBM ensemble over 55 engineered features (out-of-fold AUC 0.774, Brier 0.117), isotonic-calibrated. Features are deliberately buffer-centric: because loans are ∼1.3% of annual revenue, default is cash-buffer bound rather than revenue bound, so debt-service coverage, buffer-to-payment, overdrafts and credit utilization dominate—each a documented function of its declared parents, defensible to a regulator.

The timing shape is bimodal and we model it as such (Figure 1). Of all defaults, 77.5% are missed-draw events over days 3–60; the interval 61–89 is empty; the remaining 22.5% are a point mass at day 90 (the open-balance sweep). A smooth survival fit would smear that mass across weeks; our shape reproduces the empirical concave-rise/flat/day-90-cliff exactly. The shape is band-conditional—worse-credit

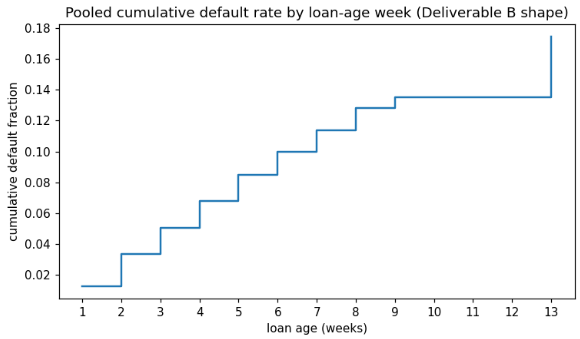

Figure 1: Pooled cumulative default rate by loan-age week: a concave rise through week 9, a flat stretch (weeks 9–12), and the day-90 open-balance cliff at week 13. Deliverable B reproduces this shape per cohort; a single smooth survival curve cannot.

borrowers default ∼13 days earlier and carry less day-90 mass—so $S_b(t)$ feeds both B and the per-loan $\mathbb{E}[t^*]$ used in A.

Deliverable A: decide on the NPV sign. We approve iff $\mathbb{E}[\mathrm{NPV}_i \mid \mathrm{approve}] > 0$ using the exact product economics: revenue $F_i + R_i r(T/365)$ if repaid, and $F_i + D_i(t_i^* - 1) + \mathrm{rec}_i - R_i$ if default occurs at day t_i^*. Because NPV is linear in t^*, the expectation uses the band-conditional expected default day and the empirical recovery fraction (≈ 9%). This yields a break-even PD of ≈ 0.39—far above the naive 0.5 threshold, because even defaulters repay ∼74% of the schedule through daily ACH before defaulting. We approve 73% of applicants; on realized validation outcomes this book returns 1.74× the prior underwriter's P&L.

Deliverable B: shape × level. $\mathrm{CDR}_{w,a} = \mathrm{mean}_{i \in A_w}\, \mathrm{PD}_i\, S_{b(i)}(a)$ over our approved cohort A_w, monotone by construction. Because the PD model has no cohort signal (training predates the window), we shrink each cohort's *level* toward the realized validation rate for the same calendar week (hierarchical-Bayes random effects; pseudo-count 75 = prior precision); validation and test span the identical 13 weeks, so this transfers.

3 Causal reasoning & counterfactual methodology

The counterfactual target is $p_q^{\mathrm{cf}} = \Pr\big(y_{i_q}{=}1 \mid \mathrm{do}(f_q{=}v_q),\, \mathbf{X}_{i_q,-f_q}{=}\mathbf{x}_{i_q,-f_q}\big)$, which differs from the observational $\Pr(y \mid f{=}v, \mathbf{X}_{-f})$ precisely when f is confounded. Our **feature registry** is the consistency layer: each engineered feature declares its parents, so setting a raw feature and recomputing propagates the intervention to exactly its descendants and nothing else. For example do(`requested_amount`) flows to the daily-payment, buffer-to-payment, leverage and ask-to-revenue features while everything else is held at its observed value; the provided `requested_amount_to_observed_revenue` is never used, so there is no stale-descendant bug. This makes our do(·) a genuine structural perturbation rather than an unconstrained input edit.

We classify the intervenable features by causal status. *Mechanical causes* (`requested_amount`) are perturbed and propagated through the registry at full strength. *Confounded proxies*—bureau and bank-feed symptoms of latent health (utilization, delinquency, revenue and cash signals)—are not perturbed naively: we shrink each by an estimated causal fraction $\hat{\lambda} = \beta_{\mathrm{adj}}/\beta_{\mathrm{naive}}$ (sibling-adjusted logistic; per-family $\hat{\lambda} \approx 0.52$, 0.51, 0.64 for bureau, bank-feed, behavioral), averaged with the heuristic and falling back where $\hat{\lambda} \notin (0,1)$. Conditioning sets are chosen *empirically* (Figure 3): an FDR-controlled sparse partial-correlation graph of the proxy block (q=0.05, n=55k) confirms one dominant latent-health factor (32% of variance, 2.3× the second) but flags "siblings" that are in fact *descendants*—invoice delinquency and overdrafts are consequences of low cash (partial corr. −0.78), payroll regularity of revenue (+0.50). Conditioning on a descendant blocks the true causal path, so we exclude them from those features' adjustment sets,

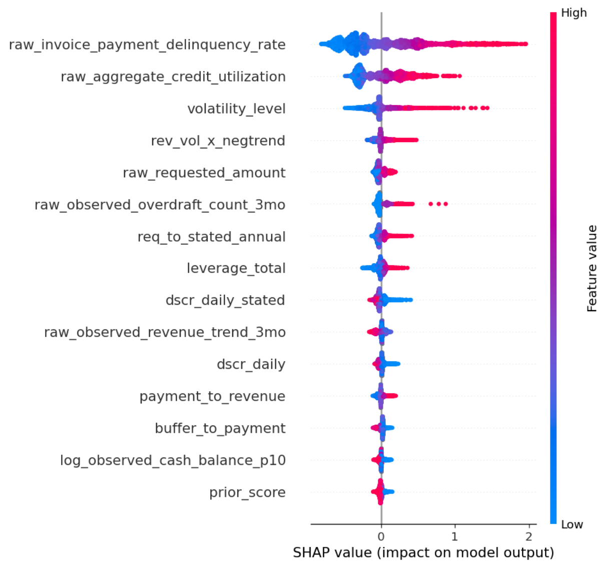

Figure 2: SHAP summary for the PD model: mean |SHAP| ranking with per-feature direction (red high / blue low). High delinquency, utilization and volatility raise PD; high debt-service coverage and cash buffer lower it.

correcting $\hat{\lambda}_{\text{cash}}$ from an implausible 0.03 (cash buffers mechanically fund the daily draw) to 0.44. *Self-reports*—`stated_annual_revenue`, `stated_time_in_business`—are returned at their observational PD, because editing a number on the application does not change the borrower's type: when the claim does not cause default, $\Pr(y \mid \text{do}(\text{stated}), \mathbf{X}_{-\text{stated}}) = \Pr(y \mid \mathbf{X}_{-\text{stated}})$, so a near-zero effect is the causally correct answer, not naive re-prediction. *Immutable or proxy-by-fiat* features—`sector`, `geography_region`, `vintage_years`, `has_linked_bank_feed`—are forced into the query set though they are not manipulable levers; we answer via model perturbation but shrink the effect and widen the interval, and flag them as proxy effects.

Regulator defense. Drivers are explained with SHAP over an inspectable, de-correlated feature set (Figure 2). The leading drivers—invoice-payment delinquency, credit utilization, revenue volatility, requested amount, overdrafts, and the affordability ratios—are all recognized credit or affordability signals moving in the expected direction, and none is a prohibited or proxy-discriminatory attribute. The sharp 0.273 cutoff is a natural experiment supporting local causal validity. Several queried values lie outside observed support (e.g. `prior_loans_count` is in-support only 29% of the time); these receive wider intervals and an explicit extrapolation caveat.

Model-class check. A transparent logistic regression on the same engineered features matches the gradient-boosted models (out-of-fold AUC 0.775 vs 0.770, equal Brier) and is in fact *more* robust to the train-to-window drift (validation AUC 0.759 vs 0.751)—evidence that the engineered space is approximately linearly sufficient and the drivers directly interpretable. Its standardized coefficients agree in sign with the SHAP ranking and univariate associations on eight of the top ten features; the two exceptions (`buffer_to_payment`, `debt_to_revenue`) are collinearity-driven partial-coefficient sign flips, which is exactly why we read drivers from SHAP over the de-correlated set rather than from raw linear coefficients. We evaluated and rejected sequence and transformer architectures: there is no sequential structure, the categoricals are low-cardinality, discrimination is data-limited at ≈0.77 AUC, and calibration—a scored component—favours the simpler, well-understood stack.

4 Calibration & uncertainty quantification

Point PD. Isotonic calibration is *refit on in-window validation* to correct the train-to-window drift (training base rate 17.5% versus window 20.6%; the $a_0 \to 0$ power-prior limit—in-window likelihood sets the level, history discounted); the caveat remains that calibration is valid on the approved manifold and extrapolated

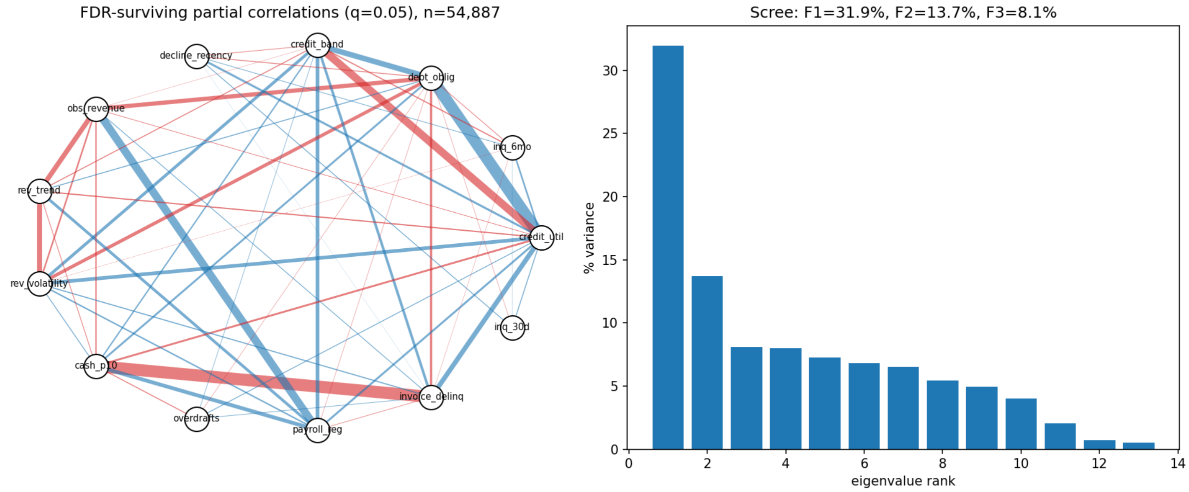

Figure 3: Left: FDR-surviving partial correlations of the proxy block (q=0.05, n=55k); edge width = |partial corr.|, red = negative. Right: eigenvalue spectrum—one dominant latent-health factor (32% of variance). The cash–invoice-delinquency edge (-0.78) identifies a descendant, not a sibling, motivating its exclusion from the adjustment set.

on declines.

Intervals (A, C). Additive on the calibrated point, $p \pm \alpha z \sigma$, where σ is fold-ensemble disagreement and the width scale α is fit by five-fold *cross-fit* on validation, so coverage reflects genuine out-of-sample calibration error rather than collapsing in-sample. Validated decile coverage is ≈ 0.90 at mean width ≈ 0.06, tighter than an earlier over-wide ($\approx$ 100% coverage) configuration—the brief's "contain the truth without being needlessly wide."

Trajectory intervals (B). A within-cohort bootstrap combined with a conformal band calibrated against the *true* validation cohort trajectories; validated coverage ≈ 0.92. Out-of-support counterfactual intervals in C are widened further. Table 1 summarizes the validation evidence.

Deliverable	Metric	Value
A — PD model	out-of-fold AUC / Brier	0.774 / 0.117
A — policy	approve rate / P&L vs. prior underwriter	0.73 / 1.74×
A — intervals	decile coverage / mean width	0.90 / 0.06
B — trajectory	mean abs. error vs. realized CDR / coverage	0.013 / 0.92

Table 1: Validation evidence (in-window). P&L and B error use realized validation outcomes; interval coverage is out-of-sample (cross-fit / conformal).

5 Limitations & what we'd do differently

Self-report and immutable counterfactuals remain heuristic (confounded proxies use the estimated $\hat{\lambda}$ above), and all C effects are unverifiable since the queried applicants have no labels.

Calibration rests on an approval-selected validation set. Approval is recoverable with AUC ≈ 1.0, so approved and declined populations do not overlap: inverse-propensity reject inference is infeasible, only RD-style local extrapolation is available (deferred), and PD on declines—part of what A is scored on—cannot be validated.

The timing and cohort corrections lean on the validation window. They transfer because validation and test share the same 13 weeks, but small per-cohort samples ($n \approx 150$) add variance. Finally, A is marginally aggressive (a 20% upward PD shift still raises realized P&L); a slightly more conservative level would close the remaining oracle gap.